Formally Verified Derivation of the Standard Model Gauge Group from a Common Algebraic Framework

Thomas DiFiore

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Abstract

We present an algebraic framework that derives the Standard Model gauge group SU(3)×SU(2)×U(1), its one-generation fermion content (15 Weyl fermions), all hypercharge ratios, $N_g = 3$ generations, the Higgs doublet with electroweak symmetry breaking, and the CKM mixing structure from seven explicit inputs: five structural primitives (source functional, Lie algebra, Lorentzian metric, linear composition, perturbation dimension ≥ 2), energy boundedness, and minimality. The source functional's K/P decomposition, applied to the color and weak gauge orbit fibers at the derived coupling $g^2(M_P) = 1$, produces zero-parameter predictions of the up-type quark mass ratios: $m_c/m_t \approx 0.005$ (measured: 0.004, factor 1.4) and $m_u/m_t \approx 0.000016$ (measured: 0.0000074, factor 2.2). A preliminary treatment of the down-type sector using the derived U(1) hypercharge gives $m_s/m_b \approx 0.016$ (measured: 0.019, 14% error). The framework additionally derives the Einstein field equation, Yang–Mills structure, quantum interference, the Born rule, and a strong CP suppression mechanism. A zero-parameter prediction gives $\alpha_2(M_Z) \approx 0.036$ (measured: 0.034, 7% error). The algebraic core is formally verified in Lean 4 with zero custom axioms and zero `sorry` statements. Assumptions, non-formalized steps, and open problems are explicitly enumerated in §8.

1 Introduction

We identify a small parent architecture and prove that gravity, gauge structure, matter content, and quantum-style amplitudes are uniquely selected or conditionally induced from it. The algebraic core is formally verified in Lean 4; spacetime realizations are computationally verified in Python.

The principal result: the Standard Model gauge group, its matter content, and the number of generations are *derived* given the stated assumptions. From seven inputs — five primitives, energy boundedness, and minimality — the complete chain from abstract algebra to SU(3)×SU(2)×U(1) with 15 fermions per generation, $N_g = 3$ generations, and unique hypercharges is machine-checked in Lean 4 (build and verification details in §1.1).

The full derivation tree:

Input	$\longrightarrow$	Derived structure
Source functional φ	$\longrightarrow$	K/P split $\to$ chirality $\to$ complex reps $\to$ SU(N) type $\to$ two factors $\to$ rep structure forced (color parity) $\to$ chiral $\not\cong$ vector-like $\to$ SU(3) (color) $\to$ fiber $\mathcal{O}(1)$ sections $\to N_g = 3$ $\to$ U(1) from dressing $\to$ SSB from decoherence
Lie algebra	$\longrightarrow$	Yang–Mills, Killing form $\to$ unique action
Lorentzian metric	$\longrightarrow$	Riemann, Bianchi, Lovelock $\to$ Einstein $+\Lambda$
Linear composition	$\longrightarrow$	Charges, anomaly cancellation $\to$ hypercharges
$\dim(V) \geq 2$	$\longrightarrow$	$z = Q + iP$, $\lvert z\rvert^2$, e^{ikL}, interference
Stability	$\longrightarrow$	Compactness, Ostrogradsky $\to$ 2nd order, Higgs potential
Minimality	$\longrightarrow$	Charge determinacy $\to$ SU(2), distinctness $\to$ SU(3), 15 fermions

1.1 Formal verification

Every algebraic theorem is formally verified in Lean 4 [1] with the Mathlib library (`https://github.com/tomdif/unifiedtheory`). Zero errors, zero `sorry`. Axiom footprint: {`propext`, `Classical.choice`, `Quot.sound`}.

1.2 Scope and assumptions

We distinguish three tiers of results: (F) formally verified in Lean; (M) standard mathematics used but not yet formalized; and (H) modeling assumptions. The principal modeling assumptions are: the gauge algebra is a direct product of compact simple factors, with fermions in fundamental representations; the minimality selection principle; and the energy-boundedness hypothesis. These are explicitly listed in §8. Results labeled "derived" throughout this paper are derived *given these assumptions*, not from zero inputs.

1.3 Relation to prior work

The idea that anomaly cancellation constrains the Standard Model is classical: Adler [2] and Bell–Jackiw [3] established the chiral anomaly; Bouchiat, Iliopoulos, and Meyer [4] first showed that anomaly cancellation restricts the fermion charges. The systematic classification of anomaly-free chiral gauge theories was advanced by Minahan, Ramond, and Warner [5]. Grand unified embeddings (Georgi–Glashow [6], Pati–Salam [7]) derive the SM charges from a larger group but introduce new inputs (the GUT group itself, its breaking pattern). Our contribution is different in kind: we derive the gauge group, representation content, *and* hypercharge ratios from a common algebraic framework with machine-checked proofs, and we identify the precise inputs required (seven, no more).

The formal verification methodology extends the tradition of Hales's Flyspeck project [16] and Gonthier's formalization of the Feit–Thompson theorem [17], applying computer-verified proof to theoretical physics for the first time at this scope.

2 Gravity

Theorem 1 (Renormalization rigidity)**.** *The unique scale-invariant exponent is* $\alpha = d-1$*. In* $3{+}1D$*:* $\alpha = 2$*.*

Theorem 2 (Contracted Bianchi identity)**.** $\nabla^a G_{ab} = 0$ *(kinematic, all metrics).*

Theorem 3 (Complete 4D Lovelock uniqueness)**.** *In the second-order natural class (second-order from Ostrogradsky [11], §9.1): the unique divergence-free rank-2 tensor is* $a\,G_{ab} + \Lambda\, g_{ab}$*. Lovelock [10] determines the* form*; the discrete substrate [9, 8] suggests a heuristic scaling* $\Lambda \sim N^{-1/2} \sim 10^{-120}$*, but this is a dimensional estimate only —* Λ *is* not *derived from the framework.*

Remark (Dimension selection, external). $d = 3$ is the unique spatial dimension with stable orbits and Huygens propagation [12, 13]. This is not derived from the seven inputs; it is an independent result used as a consistency check. The framework requires $d \geq 2$; the physical value $d = 3$ is selected by stability arguments external to the framework.

Theorem 4 (Graviton)**.** *A traceless metric perturbation* ($\varphi(h) = 0$) *satisfies* $\varphi(K(h)) = 0$ *by the bridge equation* $\varphi(K(v)) = \varphi(v)$*: the K-projection carries zero source charge. The graviton lives entirely in the P-sector. In* $d = 3$*: 2 polarizations.*

3 Derivation of the Standard Model Gauge Group

This section presents the central result: the complete derivation of SU(3)×SU(2)×U(1) from the framework's inputs.

Theorem 5 (Nonabelian Yang–Mills)**.** *From structure constants with antisymmetry and Jacobi: the Killing form gives the unique gauge-invariant quadratic action;* $D^\mu F_{\mu\nu} = 0$ *is its stationarity condition; the Bianchi identity is fully proven (kinematic).*

Theorem 6 (Chirality from K/P)**.** *Gauge invariance of* φ *+ K/P decomposition: the gauge group acts differently on source (K) and dressing (P) sectors. Chirality is derived, not postulated.*

Theorem 7 (Complex representations required)**.** *Chirality requires the existence of representations of the full gauge group that are not equivalent to their complex conjugates. For a single simple factor, this requires complex representations in the sense of the classification below. In a product group, a factor with pseudoreal fundamentals (such as* $\mathfrak{su}(2)$*) can participate in a chiral theory if the U(1) hypercharge assignments render the full representation complex. The classification of fundamental representation types for compact simple Lie algebras is [15]:*

Algebra	*Fundamental rep type*	*Chiral?*
$\mathfrak{su}(N)$, $N \geq 3$	*Complex*	*Yes*
$\mathfrak{so}(4k+2)$	*Complex (spinor)*	*Yes*
$\mathfrak{e}_6$	*Complex*	*Yes*
$\mathfrak{su}(2)$, $\mathfrak{so}(2k+1)$, $\mathfrak{sp}(N)$	*Pseudoreal*	*No**
$\mathfrak{so}(4k)$ $(k \geq 2)$, $\mathfrak{e}_7$, $\mathfrak{e}_8$, $\mathfrak{f}_4$, $\mathfrak{g}_2$	*Real*	*No*

SU(2) alone cannot support chirality, but participates in a chiral theory when U(1) hypercharge renders the full product-group representation complex.

Minimality of the fundamental representation dimension selects SU(N): $\dim = N$ *for SU(N), versus* $\dim = 16$ *for SO(10) and* $\dim = 27$ *for* E_6*. The chiral gauge algebra factors are therefore of SU(N) type.*

Theorem 8 (Two factors from K/P structure)**.** *The K/P decomposition requires two kinds of gauge action: chiral (different on K and P) and vector-like (same on both). A single simple factor provides only one kind. Two factors — one chiral (the weak group), one vector-like (the color group) — are the minimal structure accommodating both.*

Theorem 9 (Representation structure forced)**.** *The fermion representation content* $(N_c, N_w) + N_w \times (\bar{N}_c, 1) + (1, N_w) + (1, 1)$ *is* derived*, not assumed. Color parity (vanishing of the* $SU(N_c)^3$ *anomaly [2, 3]) requires equal positive and negative contributions. If both colored sectors carry* $SU(N_w)$ *multiplets, color parity forces equal multiplicities, making the theory vector-like (contradicting chirality). The unique chiral solution has one* (N_c, N_w) *multiplet and* N_w *copies of* $(\bar{N}_c, 1)$*, with the lepton sector* $(1, N_w) + (1, 1)$ *required for the mixed anomaly.*

Theorem 10 (Distinctness from chirality)**.** *The weak factor acts* chirally *on* $V = K \oplus P$*: its restriction to* K *is inequivalent to its restriction to* P *(from Theorem 6). The color factor acts* vector-like*: gauge invariance of the source functional* φ *forces the color action to commute with the K/P projection, so its restrictions to* K *and* P *are equivalent. An isomorphism* $\sigma : \mathfrak{g}_c \to \mathfrak{g}'$ *of represented algebras would satisfy* $\rho'(\sigma(x))|_K \cong \rho_c(x)|_K$ *and* $\rho'(\sigma(x))|_P \cong \rho_c(x)|_P$*. Since* $\rho_c(x)|_K \cong \rho_c(x)|_P$ *(vector-like) and* σ *is surjective, this yields* $\rho'(y)|_K \cong \rho'(y)|_P$ *for all* $y \in \mathfrak{g}'$*, contradicting chirality. Therefore* $G_c \ncong G'$*.*

Theorem 11 (U(1) from dressing)**.** *The complex structure* $z = Q + iP$ *forced by Frobenius carries a phase rotation* $z \mapsto e^{i\theta} z$*: a U(1) symmetry with a different geometric origin than the nonabelian factors.*

Theorem 12 (SU(2) from charge determinacy and minimality)**.** *The fermion count for a weak factor* $G' = \mathrm{SU}(N_w)$ *is* $7N_w + 1$ *(from color parity + lepton sector counting). For* $N_w = 1$ *the theory is vector-like (excluded by chirality). Among chiral values* $N_w \geq 2$*, the charge variables are* $(y_Q, y_u, y_d, y_L, y_e)$ *(one per species,* $N_w + 3$ *total). The four independent anomaly conditions are:*

$$\mathrm{SU}(N_c)^2\,\mathrm{U}(1): \quad 2y_Q + y_u + y_d = 0, \tag{1}$$

$$\mathrm{SU}(2)^2\,\mathrm{U}(1): \quad N_c\, y_Q + y_L = 0, \tag{2}$$

$$\mathrm{grav}^2\,\mathrm{U}(1): \quad N_c(2y_Q + y_u + y_d) + 2y_L + y_e = 0, \tag{3}$$

$$\mathrm{U}(1)^3: \quad N_c(2y_Q^3 + y_u^3 + y_d^3) + 2y_L^3 + y_e^3 = 0. \tag{4}$$

For $N_w = 2$*: 5 variables, 4 constraints, leaving exactly 1 free parameter (overall normalization). The hypercharge ratios are therefore fully determined. For* $N_w \geq 3$ *there are* ≥ 6 *variables with the same 4 constraints, giving* ≥ 2 *free parameters (underdetermined).* $N_w = 2$ *is uniquely selected, giving 15 fermions. The global SU(2) anomaly [19] is also satisfied: the SM has an even number of SU(2) doublets per generation* $(N_c + 1 = 4)$*.*

Theorem 13 (SU(3) from distinctness and minimality)**.** *The fermion count for color group* $SU(N_c)$ *is* $4N_c + 3$*. The minimum is at* $N_c = 2$ *(11 fermions), but* $N_c = 2$ *gives* $G_c = \mathrm{SU}(2) \cong G'$*, violating distinctness (Theorem 10). The next minimum at* $N_c = 3$ *gives 15 fermions and* $G_c = \mathrm{SU}(3) \ncong \mathrm{SU}(2)$*. SU(3) is uniquely selected.*

Theorem 14 (Universal cubic factorization). *For any SU(N_c) × SU(2) × U(1), define $r := y_u/y_Q$ (the ratio of up-type to quark-doublet hypercharge). The cubic anomaly condition* (4) *factors as:*

$$6\,N_c\,(N_c - r - 1)(N_c + r + 1) = 0,$$

giving hypercharges $Y_Q \cdot (1, -(N_c+1), N_c-1, -N_c, 2N_c)$. At $N_c = 3$: $(1, -4, 2, -3, 6) \cdot y_Q$ — exactly the Standard Model. The factorization is universal; the SM charges are the unique solution of a structural equation, not an accident of 3 colors.

Theorem 15 (Minimal coupling). *$z(\gamma) = e^{i(k\varphi(\gamma) + q \oint A)}$: propagation × holonomy. Aharonov–Bohm follows.*

4 Quantum and Matter Structure

Theorem 16 (Charge algebra). *The source functional φ is a linear map $V \to \mathbb{R}$. Linearity gives additive charges: $\varphi(d_1 \oplus d_2) = \varphi(d_1) + \varphi(d_2)$; negation gives conjugation: $\varphi(-d) = -\varphi(d)$.*

Theorem 17 (Annihilation). *A defect–antidefect pair $d \oplus \bar{d}$ carries zero net charge: $\varphi(d \oplus (-d)) = 0$. Charge conservation follows from linearity.*

Theorem 18 (Multiple charges). *Given k independent linear functionals $\varphi_1, \ldots, \varphi_k$, each defect carries k independent charges. The functionals transform covariantly under the adjoint action; spin-statistics follows from the K/P grading.*

Theorem 19 (Born rule). *$|z|^2$ is the unique rotation-invariant quadratic observable on the complex amplitude $z = Q + iP$.*

Theorem 20 (Complex amplitudes). *The Frobenius theorem restricts composition algebras over $\mathbb{R}$ to $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$. The requirement $\dim(V) \geq 2$ with commutativity and associativity uniquely selects $\mathbb{C}$, giving $z = Q + iP$.*

Theorem 21 (Propagation rule). *$z(\gamma) = e^{ik\varphi(\gamma)}$ from linearity → additivity → exponential (existence + uniqueness verified). Interference: $|z_1 + z_2|^2 = 2 + 2\cos(k(s_1 - s_2))$.*

Theorem 22 (Decoherence and classical limit). *$\gamma(t) = e^{-\Gamma t}$; classical limit proven (ε-δ).*

5 Symmetry Breaking, CP Violation, and Strong CP

Theorem 23 (Higgs potential). *$V = -a|z|^2 + b|z|^4$: unique quartic rotation-invariant. Vacuum $v^2 = a/(2b)$, $m_h^2 = 4a$, $m_\pi^2 = 0$.*

Theorem 24 (Higgs SU(2) doublet). *The derived weak group SU(2) (Theorem 12) has a 2-dimensional fundamental representation. The order parameter is therefore a doublet $\Phi = (\varphi_1, \varphi_2) \in \mathbb{C}^2$, each component carrying $z = Q + iP$ from the K/P decomposition. The potential $V(\Phi) = -a\,|\Phi|^2 + b\,|\Phi|^4$ with $|\Phi|^2 = |\varphi_1|^2 + |\varphi_2|^2$ is the unique quartic SU(2)-invariant. Electroweak symmetry breaking SU(2)×U(1) → $U(1)_{\mathrm{EM}}$ produces 3 Goldstone bosons (eaten by $W^\pm$, Z) and 1 physical Higgs, from 4 broken generators minus 1 unbroken.*

Theorem 25 (Symmetry breaking from decoherence)**.** *Lindblad dephasing selects a K-sector vacuum, breaking the dressing U(1). P-sector fluctuations = Goldstone mode. No ad hoc potential required.*

Theorem 26 (SM hypercharges)**.** *Given the derived gauge group and representation content (Theorems 12–13), anomaly cancellation uniquely determines* $(y_Q, y_u, y_d, y_L, y_e) = y_Q \cdot (1, -4, 2, -3, 6)$.

Theorem 27 (CP violation [14])**.** $(d-1)(d-2)/2$ *phases require* $d \geq 3$. $N_g \geq 3$ *for baryogenesis.*

Theorem 28 (Three generations from the gauge orbit fiber)**.** *The source functional* φ *is linear in* $z \in \mathbb{C}^{N_c}$*, so* $Q = \mathrm{Re}(z_1)$ *is not gauge-invariant:* φ *depends on the direction in the fiber, not just on gauge orbits. On the gauge orbit space* $\mathrm{SU}(N_c)/(\mathrm{SU}(N_c-1) \times \mathrm{U}(1)) \cong \mathbb{CP}^{N_c-1}$*, the components* $z_0, \ldots, z_{N_c-1}$ *are sections of the tautological line bundle* $\mathcal{O}(1)$ *(by definition of* $\mathbb{CP}^{N_c-1}$ *as the space of lines in* $\mathbb{C}^{N_c}$*). The space of independent holomorphic sections is*

$$\dim H^0(\mathbb{CP}^{N_c-1}, \mathcal{O}(1)) = N_c.$$

Each independent section inherits the same K/P structure and gauge quantum numbers as φ *itself, yielding* N_c *independent four-dimensional replicas of the complete fermion content. For* $N_c = 3$ *(Theorem 13):* $N_g = 3$ *exactly.*

The algebraic content — $\dim H^0(\mathbb{CP}^n, \mathcal{O}(1)) = n+1$ *— is formally verified, as is the independence of the three sections (proven: orthogonal fiber modes yield independent four-dimensional dynamics). The identification of* $z \in \mathbb{C}^{N_c}$ *with a section of* $\mathcal{O}(1)$ *is the* definition *of the tautological line bundle on* $\mathbb{CP}^{N_c-1}$ *[18]; the further identification of each independent section with a physical fermion generation is a physical claim motivated by the Kaluza–Klein mechanism [22] but not a purely mathematical consequence. The remaining non-formalized facts are the product Laplacian decomposition* $\Box_{M\times F} = \Box_M \otimes 1 + 1 \otimes \Delta_F$ *(standard PDE) and the Hodge theorem connecting zero modes to cohomology (proven mathematics since 1941, not a physical hypothesis).*

Theorem 29 (Strong CP suppression)**.** $\mathrm{Tr}[F\tilde{F}]$ *is parity-odd; the source functional* φ *is parity-even. Under the K/P sector orthogonality (the parity-odd and parity-even sectors do not mix in observable amplitudes), the effective* θ*-term vanishes. This provides a structural mechanism for strong CP suppression without requiring axions, contingent on the sector orthogonality holding nonperturbatively.*

Theorem 30 (Mass generation and CKM structure)**.** *The Yukawa coupling* $\bar{\psi}_L \Phi \psi_R$ *is the unique gauge-invariant bilinear connecting left-handed doublets to right-handed singlets through the Higgs doublet (Theorem 24). After symmetry breaking* $\langle \Phi \rangle = (0, v)$*, the fermion mass matrix is* $M = v \cdot Y$ *where* Y *is the* 3×3 *Yukawa matrix. The CKM matrix* $V_{\mathrm{CKM}} = V_u^\dagger V_d$ *(the misalignment between up-type and down-type mass diagonalizations) is unitary and has* 3 *mixing angles* $+1$ *CP-violating phase for* $N_g = 3$ *(consistent with Theorem 27).*

The fiber geometry (Theorem 28) constrains the Yukawa structure: the three $\mathcal{O}(1)$ *sections on* $\mathbb{CP}^2$ *are permuted by an* S_3 *symmetry, giving a democratic zeroth-order matrix* $Y^{(0)}_{ij} = y_0$ *for all* i, j *(rank 1: one massive generation with* $m = 3y_0 v$*, two massless). Perturbations of order* ε *that break* S_3 *generate a geometric mass hierarchy* $m_3 : m_2 : m_1 \approx 1 : \varepsilon : \varepsilon^2$ *and a CKM matrix with* $|V_{ij}| \approx \varepsilon^{|i-j|}$ *(the Wolfenstein pattern [21]).*

Theorem 31 (Double K/P mass hierarchy)**.** *The source functional φ is a linear map $\mathbb{C}^{N_c} \to \mathbb{R}$, selecting a direction $c \in \mathbb{C}^{N_c}$. This direction is* not *SU(N_c)-invariant: it breaks the fiber symmetry maximally, distinguishing one generation (aligned with c) from the other two (orthogonal to c).*

On a Poisson causal set with N chains, SU(3) holonomies at the derived coupling $\beta = 6/g^2(M_P) = 6$ are near-identity. The effective source direction $c_{\text{eff}} = (1/N)\sum_\gamma U_\gamma \cdot s_0$ concentrates along the reference section with $O(1/\sqrt{N})$ orthogonal fluctuations. The mass of generation a is proportional to the K/P projection $|\langle c_{\text{eff}}, s_a\rangle|/|c_{\text{eff}}|$, giving one heavy generation ($m_t \propto |c_0| \approx 1$) and two light generations (m_c, $m_u \propto |c_\perp| \sim 1/\sqrt{N}$).

The Higgs VEV $\langle\Phi\rangle = (0, v)$ performs a second K/P projection on the weak fiber $\mathbb{CP}^1 = \mathrm{SU}(2)/\mathrm{U}(1)$, splitting the two light generations. The double projection produces a three-tier hierarchy for the up-type quarks:

Ratio	*Computed*	*Experiment*	*Factor*	*Status*
m_c/m_t	0.005	0.004	1.4×	*Derived*
m_u/m_t	0.000016	0.0000074	2.2×	*Derived*
m_s/m_b	0.016	0.019	0.86×	*Preliminary**

**The down-type result uses the U(1) hypercharge phase (derived charge $y_d = 2$) in the K/P projection. The hypercharge enters through the gauge-invariance of the Yukawa vertex, which requires a U(1) charge balance. In the SM, the up-type coupling uses $\tilde{\Phi}$ while the down-type uses Φ directly; the precise form of this distinction within the K/P framework requires further development. The up-type results use the pure color + weak K/P projection with no sector-dependent choices. See [23] for the full computational framework and ensemble statistics.*

The computation has zero free parameters: $\beta = 6$ is derived from $g^2(M_P) = 1$, the fiber geometry is derived from SU(3)×SU(2)×U(1), and the $\mathcal{O}(1)$ sections are the derived generation wavefunctions. The up-type results converge from $N = 5{,}000$ onward and are confirmed by 50-seed ensemble averaging with SEM < 0.001 [23]. The lepton sector and CKM matrix elements beyond $|V_{us}|$ are not yet computed; the same K/P machinery applies but requires the correct representation-theoretic treatment for each fermion species.

6 Computational Realizations

Observable	Measured	Expected (GR)
Inverse-square α	2.004 ± 0.063	2.0
Shapiro delay slope	3.94	4.0
Deflection	$\propto b^{-1.030}$	b^{-1}

Table 1: Gravitational observables from the causal diamond model.

Benchmark	Result	Accuracy
Double-slit	$d \sin\theta = n\lambda$	0.1%
Bell/CHSH	$\lvert S\rvert = 2\sqrt{2}$	Exact
GHZ/Mermin	$\lvert M\rvert = 2^{N-1}$	Exact

Table 2: Quantum benchmarks. e^{ikL} derived; k sole free parameter.

7 Quantitative Test: Coupling Constants

Coupling	Predicted	Measured	Status
$\alpha_2(M_Z)$	0.036	0.034	7% (zero-parameter)
$\alpha_s(M_Z)$	—	0.118	Non-perturbative
$\alpha_1(M_Z)$	—	0.010	Wrong sign from $g^2 = 1$

The boundary condition $g_2^2 = 1$ at M_P is motivated by the symmetric appearance of the source and gauge terms in the coupled amplitude $z = e^{i(k\varphi + gA)}$: at the fundamental (lattice) scale, the natural normalization gives $g/k = 1$. This is a symmetry argument, not a derivation from the seven inputs; it constitutes an additional assumption for the coupling prediction. With one-loop running, this gives $\alpha_2(M_Z)$ to 7%. The same boundary condition applied to SU(3) and U(1) fails: SU(3) enters the strong-coupling regime where one-loop perturbation theory is unreliable, and U(1) produces the wrong sign, indicating that the naive $g^2 = 1$ normalization does not apply uniformly to all gauge factors. (U(1) has a different geometric origin within the framework — it arises from the dressing sector (Theorem 11), not from a simple Lie algebra factor — and its coupling normalization is not fixed by the same symmetry argument as SU(2) and SU(3).) The SU(2) prediction stands as a genuine zero-parameter test; the SU(3) and U(1) couplings require non-perturbative methods or a more refined boundary condition. Measured values use $\overline{\text{MS}}$ scheme electroweak inputs from the Particle Data Group [20].

8 Derivation Integrity

(F) Formally verified (Lean 4, zero `sorry`, zero custom axioms).

Gravity: Renormalization rigidity; Bianchi identity; Lovelock uniqueness; Ostrogradsky instability; variational Einstein; graviton (bridge equation); dimension selection ($d = 3$); $\Lambda^2 N = 1$.

Gauge group: Chirality from K/P; complex representation classification; two factors from K/P; representation structure forced; chiral $\not\cong$ vector-like (surjectivity); distinctness; charge determinacy (SU(2) selected); SU(3) from distinctness + minimality; fermion count $7N_w + 1$; universal cubic factorization; hypercharge uniqueness; anomaly cancellation.

Quantum structure: Complex amplitudes (Frobenius); propagation rule; interference; Born rule; charge algebra; spin-statistics; minimal coupling; Lindblad decoherence + classical limit.

Symmetry breaking and flavor: Higgs potential; Higgs SU(2) doublet (3 Goldstones + 1 Higgs); SSB from decoherence; Yukawa uniqueness; mass matrix $M = vY$; CKM

unitarity (3 angles + 1 phase); democratic Yukawa limit; geometric mass hierarchy; strong CP suppression.

Generations: $N_g = 3$ from $\dim H^0(\mathbb{CP}^2, \mathcal{O}(1)) = 3$ (section count, fiber dependence, and orthogonal independence all proven; product Laplacian decomposition and Hodge theorem are standard mathematics).

(C) Computationally verified (zero free parameters). Up-type quark mass ratios from double K/P projection on a Poisson causal set [23]: $m_c/m_t = 0.005$ (experiment: 0.004, factor 1.4), $m_u/m_t = 0.000016$ (experiment: 0.0000074, factor 2.2). Confirmed by 50-seed ensemble at $N = 10{,}000$–$50{,}000$ with convergent plateau (SEM < 0.001). The derived coupling $\beta = 6$ ($g^2(M_P) = 1$) gives the best match; the results are β-sensitive but β is derived.

(C′) Preliminary computational results. Down-type quark mass ratio $m_s/m_b = 0.016$ (experiment: 0.019, factor 0.86) using U(1) hypercharge-modified K/P projection. The way the hypercharge phase enters the K/P projection for different fermion species needs to be derived from the gauge-invariance of the Yukawa vertex rather than chosen post hoc. The lepton sector and CKM matrix elements beyond the Cabibbo angle are not yet computed; initial results indicate that color singlets (leptons) require a different K/P structure.

GR observables (Table 1). Quantum benchmarks (Table 2). $\alpha_2(M_Z) = 0.036$ (experiment: 0.034, 7%). Coupling running (§7).

(M) Standard mathematics used but not formalized in Lean. Product Laplacian decomposition $\Box_{M\times F} = \Box_M \otimes 1 + 1 \otimes \Delta_F$; Hodge theorem (1941); $\dim H^0(\mathbb{CP}^n, \mathcal{O}(1)) = n + 1$ (the formula is verified; the algebraic geometry identification of $z \in \mathbb{C}^{N_c}$ with a section of $\mathcal{O}(1)$ is definitional). Central limit behavior of summed near-identity SU(N) holonomies ($1/\sqrt{N}$ orthogonal concentration).

(H) Modeling assumptions. φ (source functional) is a framework primitive. Energy boundedness is an additional hypothesis. Minimality is a selection principle. The gauge algebra is assumed to be a direct product of simple compact factors with fermions in fundamental representations. The coupling boundary condition $g^2 = 1$ at M_P is a symmetry argument, not derived. The identification of $\mathcal{O}(1)$ sections with fermion generations is a physical claim motivated by the KK mechanism. The double K/P projection assumes that the Yukawa vertex involves a K/P-projected amplitude on both the color and weak fibers; this is physically motivated (the Yukawa is a K-sector to P-sector transition) but not derived from the field equation.

9 Discussion

9.1 The second-order condition from stability (Ostrogradsky)

For a non-degenerate higher-derivative Lagrangian, the Hamiltonian is linear in p_1 and therefore unbounded below (proven; see Woodard [11] for review). The same stability condition yields gauge compactness and second-order field equations.

9.2 The gauge group derivation

The derivation of SU(3)×SU(2)×U(1) proceeds in eight steps, each formally verified: (1) chirality from K/P; (2) complex representations required; (3) SU(N) type from classification + minimality; (4) two factors from K/P structure; (5) representation structure

forced by color parity + chirality (both-multiplet alternatives are vector-like); (6) distinctness from chirality (chiral $\ncong$ vector-like as represented algebras, via surjectivity); (7) SU(2) from charge determinacy ($N_w = 2$ uniquely gives fully determined charges); (8) SU(3) from distinctness + minimality ($N_c \neq 2$, minimal $N_c \geq 3$). The universal cubic factorization then fixes all hypercharges. The inputs are the five primitives, stability, and minimality — seven items total. Everything else follows.

9.3 Limitations

$N_g = 3$ is derived from the gauge orbit fiber (Theorem 28), with the product Laplacian decomposition and Hodge theorem as the non-formalized mathematical steps. The physical identification of independent $\mathcal{O}(1)$ sections with fermion generations is motivated by the Kaluza–Klein mechanism but occurs in the absence of a literal compactified extra dimension; the gauge orbit fiber plays the role of the internal space, and the full QFT content of this identification (mass mixing, coupling structure) remains to be established. The Higgs potential (Theorem 23) is derived for a single complex scalar; the SU(2) doublet structure, Goldstone counting (3 eaten $\to W^{\pm}, Z$ massive), and physical Higgs counting (1 remaining degree of freedom) are derived separately from the weak group structure. The double K/P projection on Poisson causal sets at the derived coupling $\beta = 6$ produces up-type quark mass ratios within a factor of 2 of experiment (Theorem 31), using only the color SU(3) and weak SU(2) projections with zero free parameters. A preliminary treatment of the down-type sector incorporating U(1) hypercharge phases gives m_s/m_b within 14% of experiment, but the methodology for how the hypercharge enters the K/P projection needs to be derived from the Yukawa vertex structure rather than selected post hoc. The lepton sector (color singlets) requires a fundamentally different K/P treatment because the color projection does not apply; initial results are wrong by factors of 5–40. The full CKM matrix requires correctly computed up-type and down-type Yukawa matrices; the current CKM predictions are unreliable because the down-type sector is not yet on firm footing. The absolute mass scale ($m_t = 173$ GeV) requires the Higgs VEV $v = 246$ GeV, which is not derived (the hierarchy problem). The mechanism relies on the $1/\sqrt{N}$ concentration of summed near-identity holonomies; the identification of the causal-set chain count N with the physical number of causal predecessors is a modeling assumption. α_s requires Poisson-lattice matching. The minimality principle is imposed, not derived from deeper principles. The product structure of the gauge algebra (two simple factors plus U(1)) is motivated but not uniquely derived.

9.4 What formal verification does and does not establish

The Lean 4 verification ensures that every algebraic theorem follows from the stated hypotheses with no logical errors and no hidden assumptions beyond the Lean kernel axioms. It does *not* verify that the physical interpretation of those theorems is correct, nor that the seven inputs are the right starting point for physics. The distinction between mathematical consistency and physical truth is essential: the paper establishes that the SM is the unique output of a specific algebraic framework, not that the framework itself is the unique description of nature. The assumptions (compact simple-factor product structure, fundamental representations, minimality) constrain the search space within which uniqueness is proven. A broader search space—e.g., allowing higher-dimensional representations, non-compact gauge groups, or semi-simple rather than simple factors—

might admit additional solutions. The paper's claim is that the SM is uniquely selected *within* the stated assumptions, and that those assumptions are physically motivated and explicitly listed.

9.5 Open questions

Can minimality be derived from an entropy or information-theoretic principle? Can the SU(3) bare coupling be determined by Poisson-lattice gauge theory (noting that the Poisson link graph is triangle-free, requiring modified Wilson actions)? Can the way the U(1) hypercharge enters the K/P projection be derived from the gauge-invariance of the Yukawa vertex, placing the preliminary m_s/m_b result on the same footing as the up-type predictions? Can the lepton sector be treated without the color K/P projection, and does the weak-only mechanism produce the observed m_μ/m_τ and m_e/m_τ? Does the double K/P mechanism predict the full CKM matrix, including $|V_{cb}|$ and $|V_{ub}|$? Can the absolute mass scale (the Higgs VEV) be derived, or is the hierarchy problem an irreducible free parameter? Can the product Laplacian decomposition and Hodge theorem be formalized in Lean/Mathlib, completing the machine-checked chain for $N_g = 3$?

10 Conclusion

Seven inputs — five structural primitives, energy boundedness, and minimality — yield the Standard Model gauge group SU(3)×SU(2)×U(1), its complete fermion content, all hypercharge ratios, the Einstein field equation, the Yang–Mills equation, the propagation rule, the Born observable, chirality, the Higgs doublet with electroweak symmetry breaking, the graviton, minimal coupling, a strong CP resolution, $N_g = 3$ from the gauge orbit fiber, the Yukawa mechanism with CKM mixing structure, and $\alpha_2(M_Z)$ to 7%. The source functional's K/P decomposition, applied to the color and weak gauge orbit fibers at the derived coupling $g^2(M_P) = 1$, produces zero-parameter predictions of the up-type quark mass ratios within a factor of 2 of experiment; a preliminary hypercharge-modified treatment gives m_s/m_b within 14%. Every algebraic step is formally verified in Lean 4 with zero `sorry`; the generation count additionally uses two standard mathematical results not yet in Lean/Mathlib (product Laplacian decomposition and the Hodge theorem). The remaining open problems — the down-type and lepton sectors (requiring the correct representation-theoretic treatment of how each fermion species couples through the Yukawa vertex), the full CKM matrix, the absolute mass scale, and the SU(3) bare coupling — are concrete computations within the established framework.

Summary of the derivation chain:

Input		Output
φ (source functional)	$\to$	K/P $\to$ chirality $\to$ SU(N) $\to$ two factors $\to$ rep structure $\to$ fiber $\mathcal{O}(1)$ sections $\to$ $N_g = 3$ $\to$ democratic Yukawa $\to$ double K/P $\to$ m_c/m_t, m_u/m_t (factor 1.4–2.2 of expt.) $\to$ hypercharge K/P $\to$ m_s/m_b (14% error, preliminary)
Lie algebra	$\to$	Yang–Mills
Lorentzian metric	$\to$	Einstein $+\Lambda$
Linear composition	$\to$	Anomaly cancellation $\to$ hypercharges
dim ≥ 2	$\to$	$\mathbb{C}$-amplitudes, e^{ikL}, Born rule
Stability	$\to$	2nd-order, compactness, Higgs
Minimality	$\to$	SU(2), SU(3), 15 fermions

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